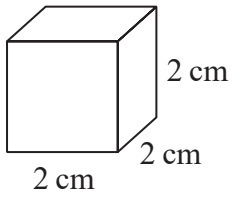
12 The diagram shows a cube of side length 2 cm.



Vera says,

"The volume of any solid made with 6 of these cubes is 48 cm^3"

(a) Is Vera correct?
You must show your working.

One cube: $2 \times 2 \times 2 = 8\ cm^3$

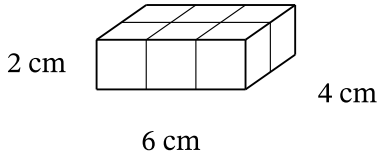
1 mark
Find the volume of one cube

Yes. Six cubes have volume $6 \times 8 = 48\ cm^3$.

1 mark
Yes, supported by 48 cm^3

(2)

(b) (i) Draw a cuboid that can be made with 6 of these cubes.
Write the dimensions of the cuboid on your diagram.



1 mark
Cuboid with lengths
6 cm, 4 cm and 2 cm

$3 \times 2 \times 1$ cubes, each with 2 cm edges.

(1)

(ii) Work out the surface area of your cuboid.

Two 6×4 faces: $2 \times 24 = 48\ cm^2$

Two 6×2 faces: $2 \times 12 = 24\ cm^2$

Two 4×2 faces: $2 \times 8 = 16\ cm^2$

Total $= 48 + 24 + 16 = 88\ cm^2$

1 mark
Find and add the face areas

88 cm^2

1 mark
Correct surface area

(2)

(Total for Question 12 is 5 marks)